The bigger picture of linear transformations and linear algebra

Definition: Linear Combination.
Let V be a Vectorial space over $\mathbb{F}$, $c_1, c_2, c_3, ..., c_n \in \mathbb{F}$ and $v_1, v_2, v_3, ..., v_n \in V$. A linear combination of vectors in V is defined as follows:

$$\sum_{i=1}^{n} c_i v_i = c_1 v_1 + c_2 v_2 + c_3 v_3 + ... + c_n v_n$$

Definition: Span.
Let $S = \{v_1, v_2, ..., v_k\}$ be a set of vectors over V, $\text{span}(S)$ is defined as every linear combination of vectors in S.

$$\text{span}(S) \coloneqq \{c_1 v_1 + c_2 v_2 + c_3 v_3 + ... + c_n v_n \mid c_1, c_2, c_3, ..., c_n \in \mathbb{F}\}$$

Intuitive defiinition: $\text{span}(S)$ in say, $\mathbb{R}^2$ is the set of every point that you can reach using vectors in S and scalars in $\mathbb{R}$, i.e a "plane" bounded by vectors in S.

Definition: Linear independant vectors.
Let $u, v \in V$. There are several ways to tell that u and v are linearly independant, intuitively, they are not colinear or coplanar and therefor $\text{span}(u, v) \neq \varnothing$. But formally:
- $\det\left(\begin{pmatrix} u \\ v \end{pmatrix}\right) \neq 0$
- $c_1 v_1 + c_2 v_2 + ... + c_n v_n = 0 \Rightarrow c_1 = c_2 = ... = c_n = 0$
- there exists no solution to $u \cdot c = v$.

Definition: Basis.
Let $u, v \in V$. u and v are basis if:
- u and v are linearly independant
- $\forall w \in V \quad \exists c_1, c_2 \in \mathbb{F} \mid w = uc_1 + vc_2$

Intuitive defiinition: Every vector in V can be generated as a linear combination of u and v.

Important results on linear transformations

The rows of a square matrix can be interpreted as the basis vectors.

If we have a vector (x, y) in a vectorial space ($\mathbb{R}^2$ in this purpose) and we linearly transform it using a square matrix. We can express this as:

$$(x, y)\begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} = x(m_{11}, m_{12}) + y(m_{21}, m_{22}).$$

I.e the result of the transformation is a linear combination of the rows in the matrix and the scalars x, y in (x, y). This result is important because this means:

$$x(m_{11}, m_{12}) + y(m_{21}, m_{22}) \in \text{span}((m_{11}, m_{12}), (m_{21}, m_{22})).$$

Furthermore, if we want to assure the transformation could reach every point and therefore be a succesful linear transformation on $\mathbb{R}^2$, then $(m_{11}, m_{12}), (m_{21}, m_{22})$ must be both basis in $\mathbb{R}^2$.

To transform a vector from the original coordinate space to the new coordinate space, we multiply the vector by the matrix.